

Rishik Kasula

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SUMMARY

AI/ML Engineer building and testing Python backend, workflow, and automation systems; M.S. in Computer Science (Data Science), UNC Charlotte (May 2026, GPA 3.80). Author of three public, CI-tested projects: a streaming data-quality gate for Apache Kafka, a calibrated card-fraud scoring service, and an XBRL-graded LLM extraction harness.

TECHNICAL SKILLS

Machine Learning & AI: Python, scikit-learn, Classification, Feature Selection, Model Evaluation, Probability Calibration, LLM Evaluation Harnesses, Retrieval-Grounded Generation, OpenAI API

Backend & Automation: FastAPI, REST APIs, Workflow State Machines, Idempotent APIs, Google Apps Script, Google Sheets, Slack API

Data Engineering: Apache Kafka, ETL, SQL, Change Data Capture (CDC), Data Quality & Validation, Avro, AWS Glue/S3, DuckDB

Engineering & Tooling: Docker, CI (GitHub Actions), Git, Code Review, pytest, unittest, mypy, Power BI, Linux

EDUCATION

University of North Carolina, Charlotte

Charlotte, NC

Master of Science (M.S.) in Computer Science — Data Science Concentration

GPA: 3.80 | Graduated May 2026

Coursework: Artificial Intelligence, Big Data Analytics, Knowledge Discovery in Databases, Visual Analytics, Database Systems

Vellore Institute of Technology (VIT-AP)

Amaravati, India

B.Tech in Computer Science (Networking & Security)

CGPA: 8.31/10 | May 2024

Coursework: Data Structures, DBMS, Object-Oriented Programming, Cloud Computing, Foundations for Data Analytics

WORK EXPERIENCE

AI/ML Engineer

Remote

Rebecca Everlene Trust Company

Aug 2026 – Present

- Closed a workflow-state gap in a Python/FastAPI participant onboarding and allocation system that left accepted candidates invisible to task allocation; added a guarded, idempotent status-transition endpoint that rejects invalid transitions (HTTP 409) with zero partial writes, plus audit-history logging and regression tests.
- Peer-reviewed a teammate's email-idempotency change and identified reliability defects — failed sends recorded as delivered, duplicate notifications, and unsafe retry state across multiple recipients; all were resolved before merge.
- Hardened a Slack automation built on Google Apps Script ahead of rollout: moved credentials out of active source into a managed configuration store, added an explicit test/production run mode and code-controlled scheduling, and validated paginated Slack user retrieval.

Graduate Research Assistant & Instructional Assistant

Charlotte, NC

UNC Charlotte, College of Computing & Informatics

Sep 2025 – May 2026

- Designed and integrated an information-theoretic feature-weighting module into the composite scoring layer of *RefExp*, a faculty-led static-code-analysis research tool built on Joern code property graphs.
- Automated the tool end-to-end with a single-command pipeline runner and terminal reporting, and fixed 3 query and scoring bugs affecting 60+ evaluation rows.
- As Instructional Assistant for a graduate Software Engineering for AI-Enabled Systems course, mentored student teams across the ML lifecycle — data collection, cleaning, labeling, feature engineering, training, and evaluation.

Intern

India

Inspirisys Solutions Ltd.

Jan 2024 – May 2024

- Engineered an Apache Kafka + ZooKeeper ingestion pipeline for real-time querying over banking-IT data, applying ETL, Change Data Capture (CDC), and PII handling.
- Deployed PrivateGPT in both local open-source-model and OpenAI-API configurations on a Windows/CUDA environment using Hugging Face models.
- Developed Power BI dashboards tracking banking-IT KPIs.

PROJECTS

tickflow — Streaming Data-Quality Gate for Kafka-Compatible Streams (v0.9)

github.com/RISHIKKASULA/tickflow

- Built and released (v0.9, MIT, pre-1.0 with a published roadmap) an open-source contract-enforcement layer for Apache Kafka streams: per-topic Avro contracts and semantic rules enforced inline, violations quarantined with self-describing envelopes; 194 tests at 99.4% core-module coverage, mypy strict, CI on every push.
- In a seeded fault-injection benchmark, replaying with the gate on vs. off cut violated bars from 1,076 to 0 of 15,061, with gated output bit-identical to the ground-truth projection (fixture-scale).

fraudscore — Calibrated Card-Fraud Scoring Service

github.com/RISHIKKASULA/fraudscore

- Shipped a Python service (scikit-learn, FastAPI, DuckDB, Docker, CI) that prices every fraud decision — reviewing a transaction when $P(\text{fraud}) \times \text{amount}$ exceeds the review cost.
- Under leak-free chronological evaluation, an amount-aware rule saved \$84.87 per 10k transactions (~13% cheaper than the best single-threshold baseline); a plain logistic model beat gradient boosting.

filingslens — Evaluation Harness for LLM Extraction from SEC Filings

github.com/RISHIKKASULA/filingslens

- Released an evaluation harness that grades local-LLM financial-figure extraction against each company's own XBRL filings — a 600-call deterministic grid with cluster-bootstrap confidence intervals; mypy strict, coverage-gated CI.
- Measured a model-size cliff on headline-figure extraction with statements in context (N=10 large-cap filers): an 8B local model reached 87.8%, well clear of two smaller models whose confidence intervals do not separate; hand-labeled 230 failures into a six-label taxonomy.

CERTIFICATIONS & OPEN SOURCE

Certifications: AWS Academy Data Engineering (Apr 2025, 40 hrs); DeepLearning.AI Mathematics for ML & Data Science (Feb 2025); IBM Python for Data Science, AI & Development; IBM/Cognitive Class Docker Essentials (Apr 2025)

Open Source: Merged a bug fix in py-pdf/pypdf (10.1k GitHub stars) correcting outline-item tree removal; full patch coverage, two maintainer review rounds.